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Web Technology

Quality Laboratory Manual

ExperimentNo. 10

Design web page to perform CRUD operation



Course Instructor–

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Experiment No.10

Title of Experiment: Design web page to perform CRUD operation

Aim of Experiment: To design and implement a web application that performs CRUD (Create, Read, Update, Delete) operations using HTML, CSS, PHP, and MySQL.

System Requirements –

- Operating System: Windows 10 / Linux / macOS
- RAM: Minimum 4 GB
- Processor: Intel i3 or above

Software/s Needed for Experiment-

- Web Browser (Chrome / Firefox)
- Code Editor (VS Code / Sublime Text)
- XAMPP / WAMP / LAMP Server
- PHP 7.x or above
- MySQL Database

Experiment Objectives:

- To understand CRUD operations in web applications
- To implement database connectivity using PHP and MySQL
- To design forms for inserting and updating data
- To retrieve and display data from database
- To delete records securely from database

Experiment Outcomes:

- Develop a dynamic web application using PHP and MySQL
- Perform Create, Read, Update, Delete operations
- Understand client-server interaction
- Implement database-driven web pages
- Gain practical knowledge of backend development

Theory:

CRUD operations are the basic functions used to interact with a database:

- **Create:** Insert new records into database
- **Read:** Retrieve and display records
- **Update:** Modify existing records
- **Delete:** Remove records from database

In web applications, CRUD operations are implemented using server-side languages like PHP and database systems like MySQL. Forms are used to collect user input, and SQL queries are used to manipulate the database.

Program

1. Database Creation (MySQL)

```
CREATE DATABASE crud_db;

USE crud_db;

CREATE TABLE students (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(50),
    email VARCHAR(50)
);
```

2. db.php (Database Connection)

```
<?php
$conn = mysqli_connect("localhost", "root", "", "crud_db");

if(!$conn){
    die("Connection Failed: " . mysqli_connect_error());
}
?>
```

3. index.php (Display Records)

```
<?php include 'db.php'; ?>

<!DOCTYPE html>
<html>
<head>
    <title>CRUD Operation</title>
</head>
<body>

<h2>Student Records</h2>

<a href="insert.php">Add New Student</a><br><br>

<table border="1">
<tr>
    <th>ID</th>
    <th>Name</th>
    <th>Email</th>
    <th>Action</th>
</tr>

<?php
$result = mysqli_query($conn, "SELECT * FROM students");

while($row = mysqli_fetch_assoc($result)){
?>
<tr>
    <td><?php echo $row['id']; ?></td>
    <td><?php echo $row['name']; ?></td>
    <td><?php echo $row['email']; ?></td>
    <td>
        <a href="edit.php?id=<?php echo $row['id']; ?>">Edit</a> |
        <a href="delete.php?id=<?php echo $row['id']; ?>">Delete</a>
    </td>
</tr>
</tr>
```

```
<?php } ?>
```

```
</table>
```

```
</body>
```

```
</html>
```

4. insert.php (Create Operation)

```
<?php include 'db.php';
```

```
if(isset($_POST['submit'])){
    $name = $_POST['name'];
    $email = $_POST['email'];

    mysqli_query($conn, "INSERT INTO students(name, email)
VALUES('$name','$email')");
    header("Location: index.php");
}
?>
```

```
<form method="post">
Name: <input type="text" name="name" required><br><br>
Email: <input type="email" name="email" required><br><br>
<input type="submit" name="submit" value="Add">
</form>
```

5. edit.php (Update Operation)

```
<?php
```

```
include 'db.php';
```

```
$id = $_GET['id'];
$result = mysqli_query($conn, "SELECT * FROM students WHERE id=$id");
$row = mysqli_fetch_assoc($result);

if(isset($_POST['update'])){
    $name = $_POST['name'];
    $email = $_POST['email'];

    mysqli_query($conn, "UPDATE students SET name='$name', email='$email' WHERE
id=$id");
    header("Location: index.php");
}
?>
```

```
<form method="post">
Name: <input type="text" name="name" value="<?php echo $row['name'];
?>"><br><br>
Email: <input type="email" name="email" value="<?php echo $row['email'];
?>"><br><br>
<input type="submit" name="update" value="Update">
</form>
```

6. delete.php (Delete Operation)

```
<?php
```

```
include 'db.php';
```

```
$id = $_GET['id'];
```

```
mysqli_query($conn, "DELETE FROM students WHERE id=$id");  
  
header("Location: index.php");  
?>
```

Task for Students: Students will design and implement a web-based CRUD application using PHP and MySQL.

Task Requirements:

- Create database and table
- Implement insert functionality
- Display records in table format
- Implement update feature
- Implement delete functionality
- Ensure proper navigation between pages

Submission Requirement:

- PHP Source Code (All Files)
- Screenshot of Insert Operation
- Screenshot of Display Page
- Screenshot of Update Operation
- Screenshot of Delete Operation
- Brief Report (Explanation + Working Flow)

Observations:

The experiment demonstrated how CRUD operations are implemented in a web application. Data was successfully inserted, retrieved, updated, and deleted using PHP and MySQL. The system provided dynamic interaction between user interface and database.

Conclusion:

CRUD operations are fundamental to web development. This experiment helped in understanding database handling, server-side scripting, and building dynamic web applications.

Expected Oral Questions:

1. What is CRUD?
2. What is primary key?
3. Difference between GET and POST?
4. What is SQL injection?
5. How to connect PHP with MySQL?
6. What is mysqli?

FAQs in Interview:

1. What are prepared statements in PHP?
2. How to secure CRUD operations?
3. Difference between MySQL and MySQLi?
4. What is normalization?
5. How to prevent SQL injection?
6. What is MVC architecture?